

SANYA PHOENIX INTL, China

WMO: 574941

Lat: 18.303N

Lon: 109.412E

Elev: 28

StdP: 100.99

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	16.1	7.8	6.6	17.9	9.9	7.6	18.8	8.9	25.9	8.3	25.2	2.5	80	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.9	33.1	27.4	32.2	27.1	32.1	27.0	28.6	31.5	28.2	31.0	27.9	30.7	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.1	31.2	27.2	23.0	30.7	27.1	22.9	30.6	92.7	31.5	91.2	31.9	90.0	31.0	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.3	DB	11.7	34.2	1.7	0.9	10.5	34.8	9.5	35.3	8.5	35.8	7.3	36.4	
			WB	8.8	29.1	1.4	0.5	7.7	29.4	6.9	29.7	6.1	30.0	5.1	30.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	22.3	23.1	25.0	27.3	28.9	29.3	28.6	28.4	27.9	26.8	25.5	23.2
	DBStd	2.93	2.32	2.39	2.11	1.35	1.31	1.27	1.30	1.15	1.09	1.30	1.74	2.36
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	7	2	2	1	0	0	0	0	0	0	0	0	2
	CDD10.0	5980	381	367	465	519	586	579	577	572	538	521	466	409
	CDD18.3	2944	125	135	207	269	328	329	318	313	288	263	216	153
	CDH23.3	29613	536	720	1525	2759	4067	4256	3869	3677	3173	2464	1733	833
	CDH26.7	10309	58	94	297	844	1730	1924	1548	1378	1104	745	443	143
(14) Wind	WSAvg	3.0	3.5	3.3	3.3	2.9	2.6	2.4	2.6	2.6	2.6	3.1	3.2	3.4
(15) Precipitation	PrecAvg	1767	17	19	30	78	172	183	235	265	312	307	106	44
	PrecMax	2174	72	78	105	181	400	371	393	524	597	928	326	186
	PrecMin	1192	0	2	2	3	40	72	127	41	125	38	4	3
	PrecStd	273	15	17	29	45	85	69	68	97	139	198	85	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.8	30.8	32.2	33.9	33.8	33.2	33.1	33.0	32.9	31.8	30.2
		MCWB	22.1	23.0	24.2	26.0	27.3	27.7	27.5	27.7	26.8	25.0	24.5	23.5
	2%	DB	28.1	28.8	29.8	31.2	32.8	33.1	32.2	32.2	32.1	31.9	30.9	29.1
		MCWB	21.9	23.0	24.4	26.0	27.3	27.7	27.4	27.4	26.9	24.9	24.4	22.7
	5%	DB	27.1	27.9	29.0	30.8	32.0	32.2	32.0	31.9	31.2	31.0	30.0	28.1
		MCWB	21.6	22.7	24.1	25.9	27.0	27.4	27.4	27.3	26.8	24.9	24.0	22.1
	10%	DB	26.1	27.0	28.2	30.0	31.2	32.0	31.2	31.1	31.0	30.1	29.1	27.1
		MCWB	20.8	22.2	23.6	25.5	26.9	27.3	27.1	27.1	26.7	24.6	23.6	21.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	24.6	25.7	27.2	28.8	29.1	29.0	28.9	28.3	27.0	26.4	24.6
		MCDB	26.9	27.9	29.2	30.2	31.7	32.1	32.0	31.9	31.0	30.5	30.2	28.9
	2%	WB	23.4	24.0	25.2	26.7	28.2	28.6	28.2	28.2	27.9	26.5	25.6	23.8
		MCDB	26.3	27.2	28.7	29.9	31.0	31.6	31.0	31.0	30.6	29.8	29.2	27.7
	5%	WB	22.5	23.5	24.7	26.4	27.8	28.2	27.9	27.7	27.3	26.0	25.0	23.1
		MCDB	25.7	26.7	28.1	29.6	30.7	31.1	30.7	30.6	30.3	29.1	28.5	26.7
	10%	WB	21.8	22.9	24.2	26.0	27.4	27.8	27.4	27.2	26.9	25.5	24.5	22.4
		MCDB	25.2	26.1	27.5	29.2	30.4	30.7	30.4	30.2	29.9	28.6	27.8	26.1
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	6.7	6.1	5.6	5.1	4.9	4.9	5.2	5.8	7.1	7.4	7.6
		MCDBR	7.6	7.2	6.2	5.9	5.3	5.1	5.5	5.9	6.4	8.1	8.0	8.2
		MCWBR	3.5	3.3	2.9	2.7	2.6	2.6	2.9	3.0	3.2	3.5	3.5	3.7
	5% WB	MCDBR	6.5	5.9	5.4	5.1	4.7	4.9	5.4	5.8	6.3	6.7	7.5	7.5
		MCWBR	3.3	3.1	2.8	2.7	2.6	2.7	2.9	3.0	3.2	3.1	3.4	3.7
(40) Clear-Sky Solar Irradiance	taub		0.505	0.525	0.629	0.593	0.484	0.470	0.451	0.477	0.532	0.569	0.510	0.529
	taud		2.022	2.006	1.781	1.921	2.275	2.329	2.379	2.291	2.098	1.941	2.079	1.966
	Ebn at Noon		772	782	716	746	822	826	843	828	782	734	762	733
	Edh at Noon		164	175	226	198	137	129	123	135	162	184	154	169
(44) All-Sky Solar Radiation	RadAvg		3.67	4.28	4.66	5.47	5.78	5.42	5.34	5.08	4.78	4.50	4.03	3.46
	RadStd		0.46	0.60	0.35	0.43	0.40	0.45	0.52	0.36	0.49	0.54	0.33	0.34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (5 neighbors)	+0.26	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	+90	+83

Nomenclature: See separate page